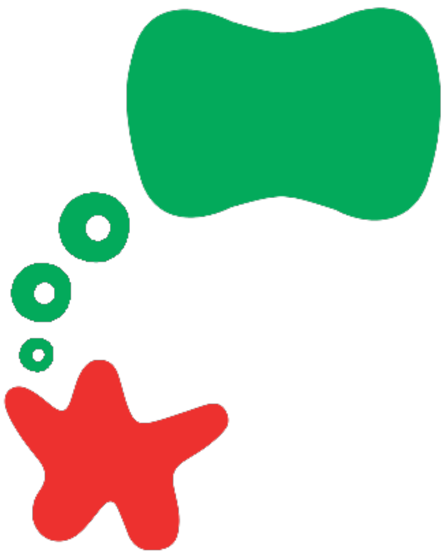
ThoughtWorks®

# User Stories101

# What is a User Story?



# User stories

- A tool for **iterative** development
- Represents a **unit** of work that should be developed
- Helps **track** that piece of functionality's lifecycle
- It is a token for a **conversation**, a placeholder for a **conversation**

# Why should I use them?

- It is a piece of customer-visible functionality written in common language

  *Universally understood*

- Ensure you only build things for a reason

  *Prevents waste*

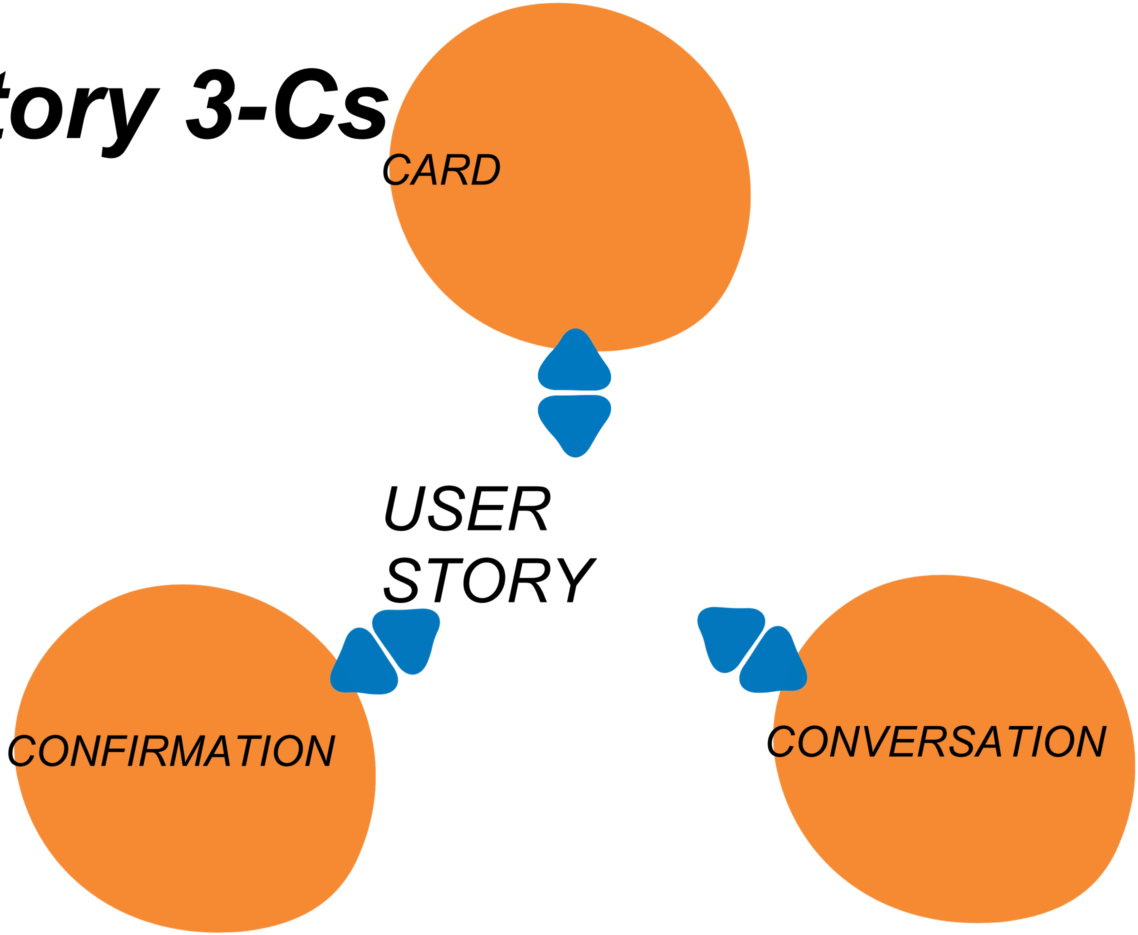
- Simple and flexible

  *Minimum overhead*

- Proven way of gathering requirements on agile projects

  *Effective teams*

# Story 3-Cs



# Story 3-Cs

CARD

USER STORY

CONFIRMATION

CONVERSATION

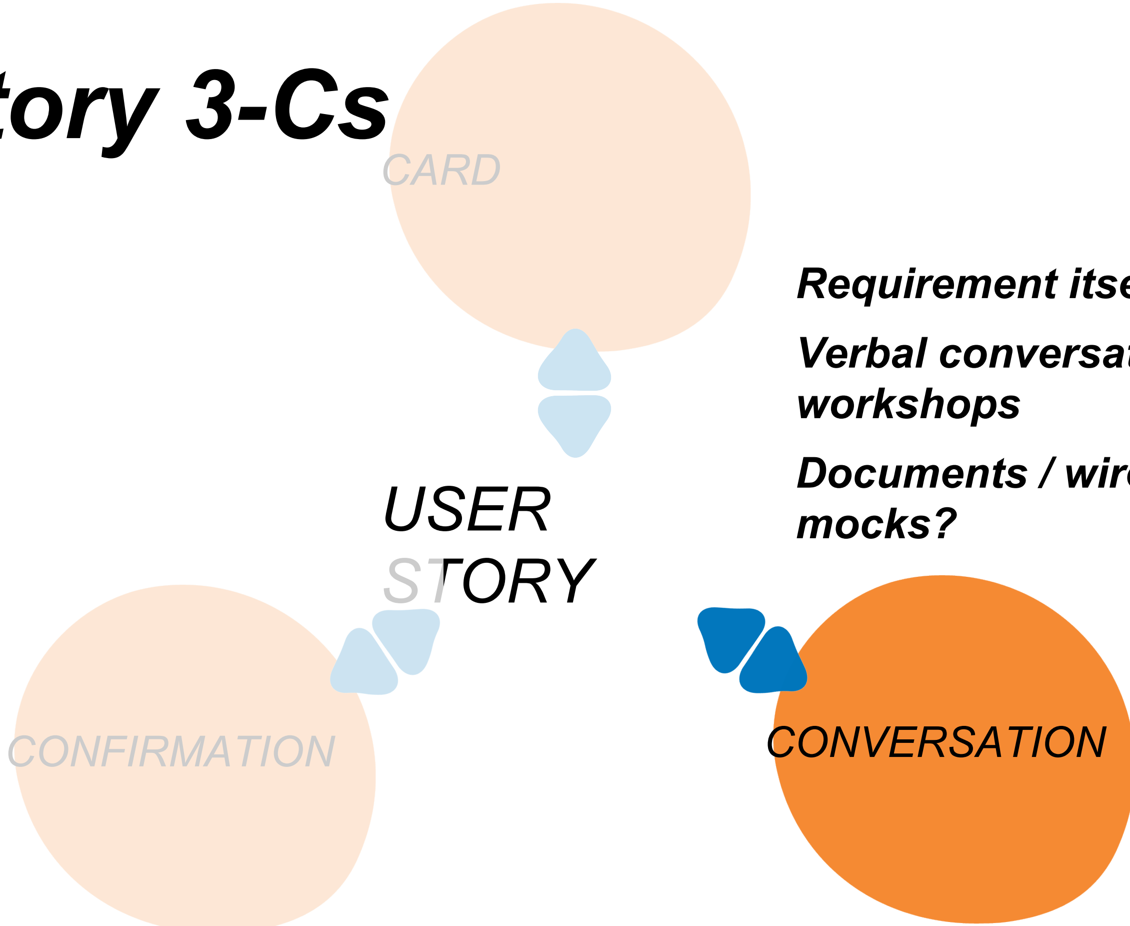
***Physical token***

***Used in planning***

***Reminder for a conversation***

***Often annotated***

# Story 3-Cs




***Requirement itself***

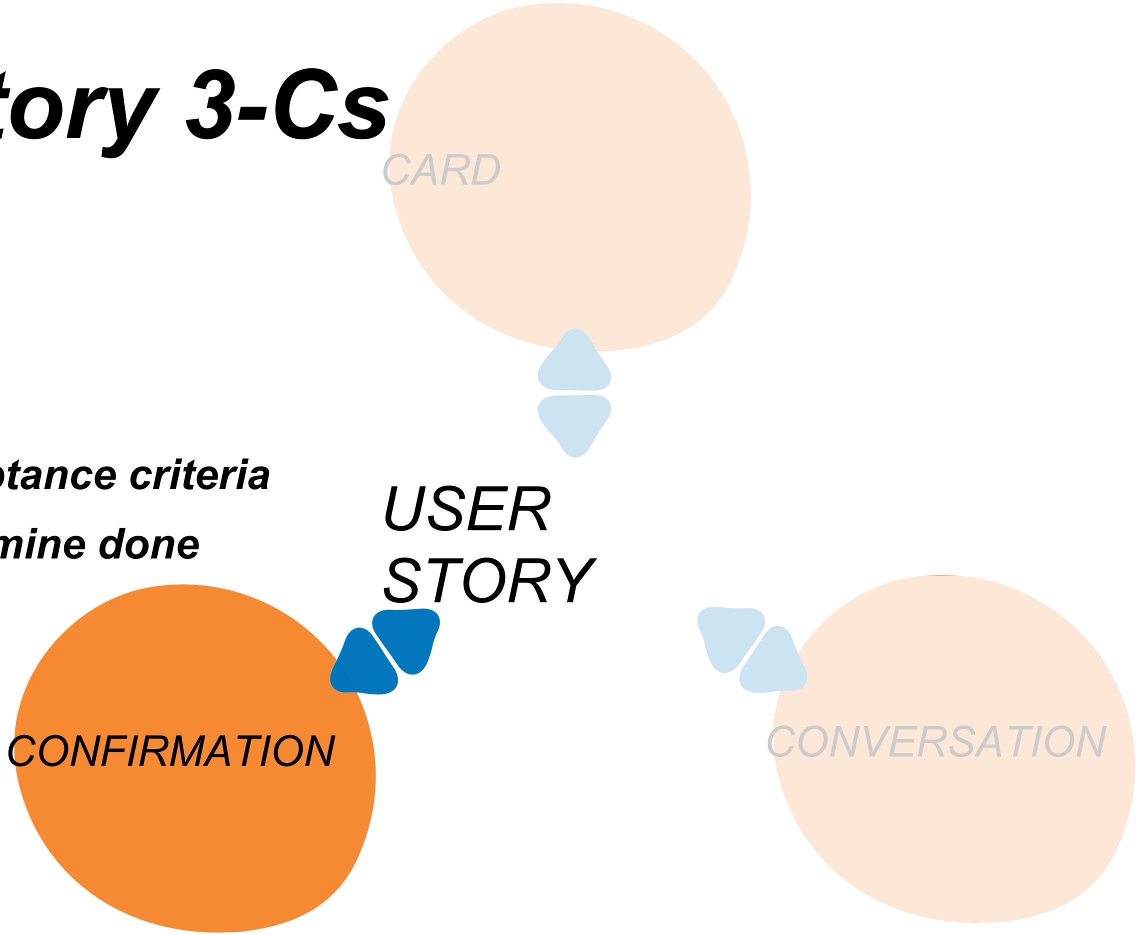
***Verbal conversation / workshops***

***Documents / wireframes / mocks?***

# Story 3-Cs

***Acceptance criteria***

***Determine done***



# In a nutshell

Moving from statements about what the system should do…

*The system shall <do something>*

to a concise description of a piece of functionality that will be valuable to a user (or owner) of the software

*As a <role>,*
*I want to <business goal>,*
*so that <value/motivation>*

ThoughtWorks®

As a <role>

I want to <business goal>

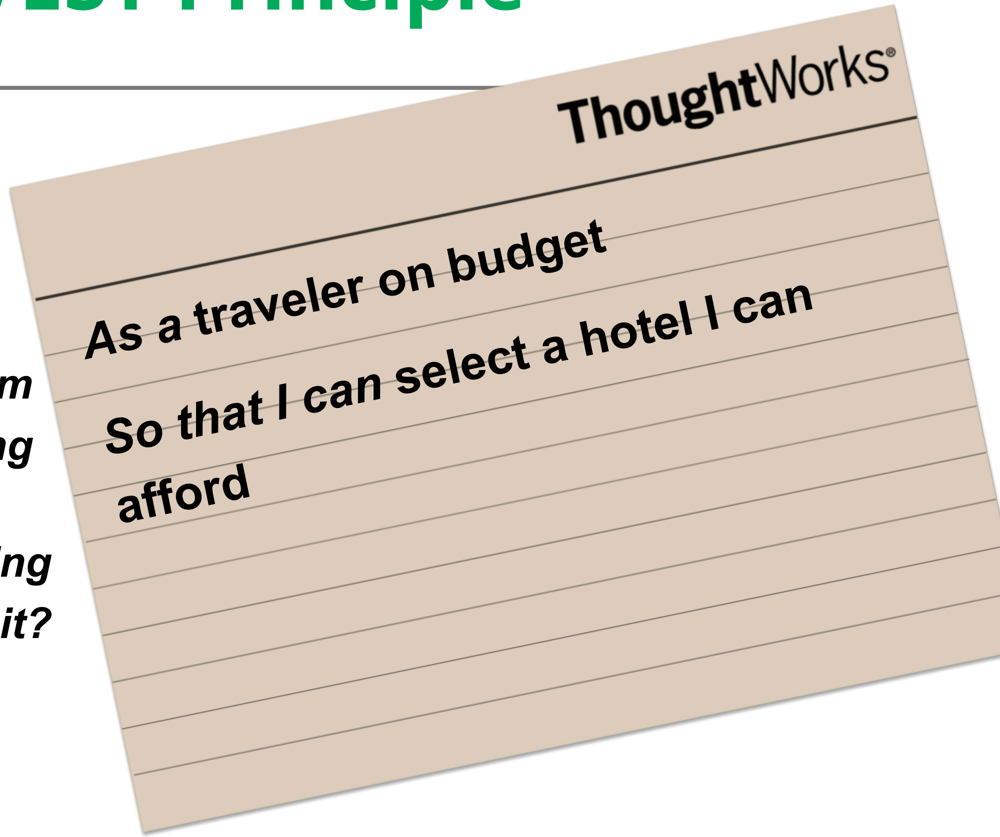
So that <value>

# INVEST Principle

| | | |
|---|---|---|
| **I** | Independent | *No overlap – order is ok!* |
| **N** | Negotiable | *No contract. Details can change.* |
| **V** | Valuable | *Incremental benefit to something.* |
| **E** | Estimable | *Relative size to other stories.* |
| **S** | Small | *Shouldn't be bigger than an iteration.* |
| **T** | Testable | *Should be able to tell when it is done.* |

# INVEST Principle

**V** Valuable

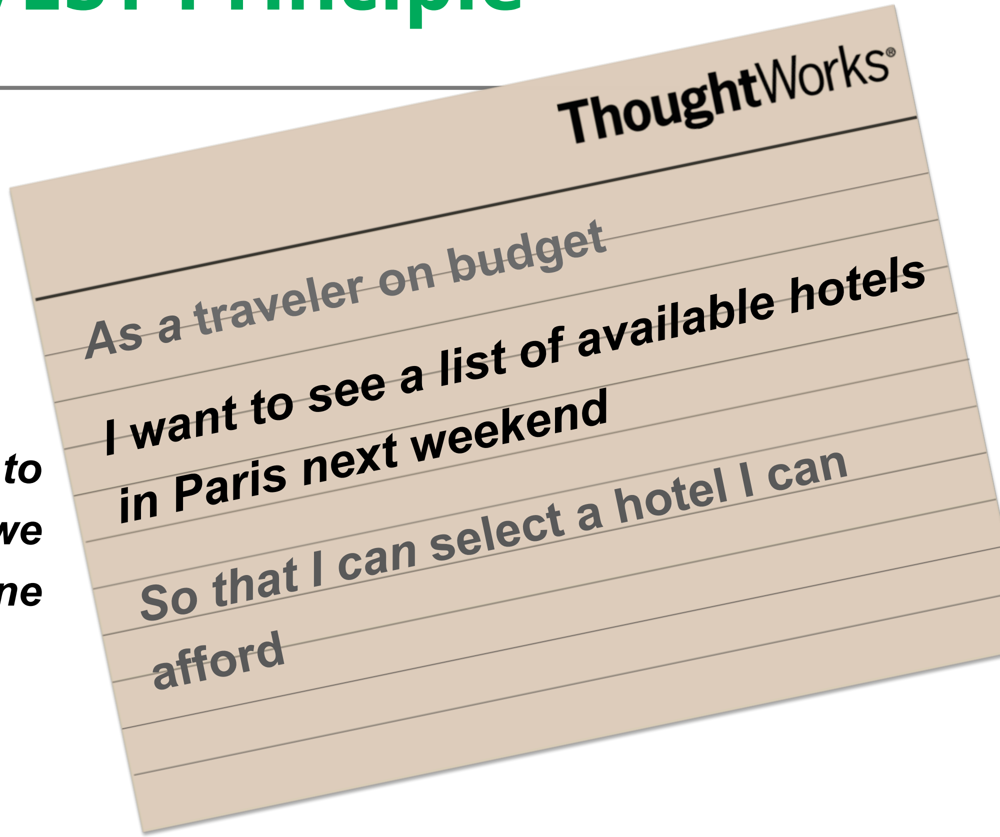



***Someone is benefiting from what we are building***

***Why are we building it?***

# INVEST Principle

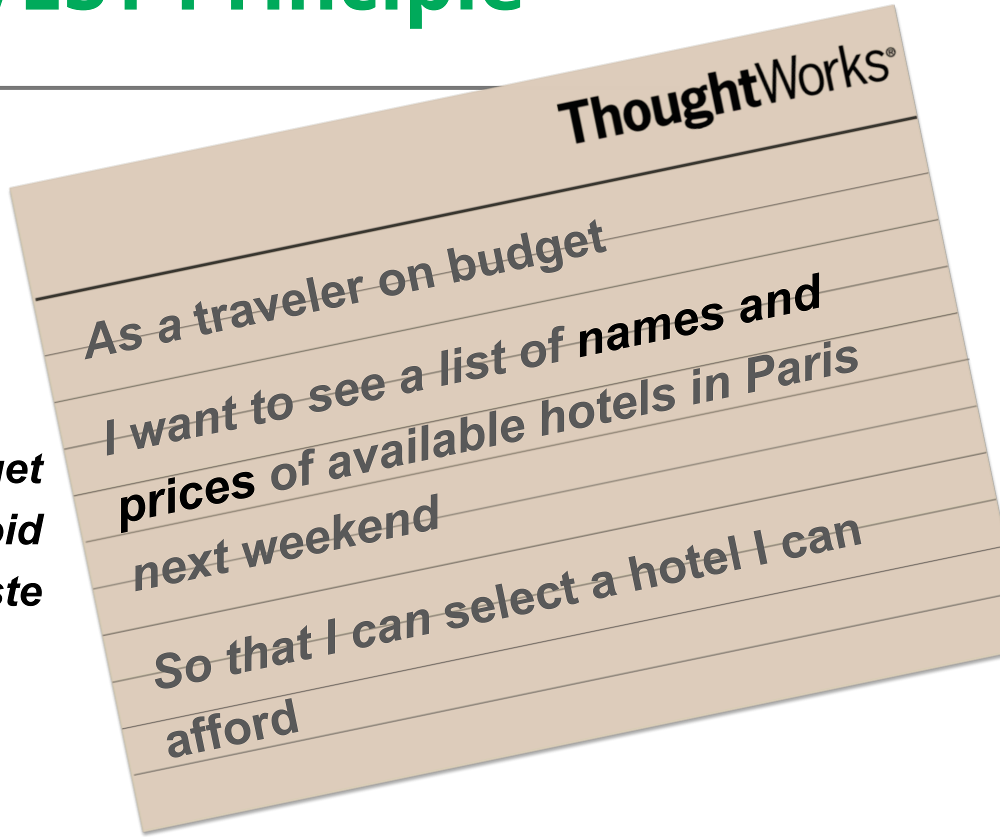
**T** Testable




***Now we know what to do and know when we will be done***

# INVEST Principle

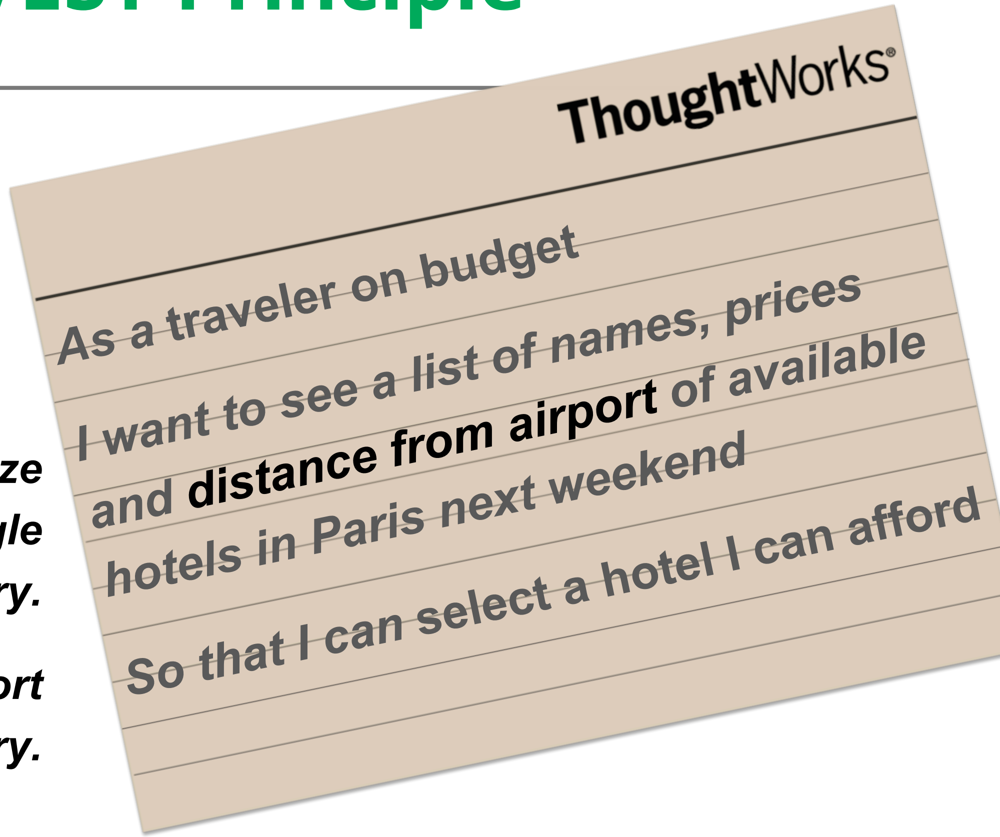
**S** Small




***Just enough to get feedback and avoid waste***

# INVEST Principle

**I** Independent




***Should be able to realize the value of each single story.***

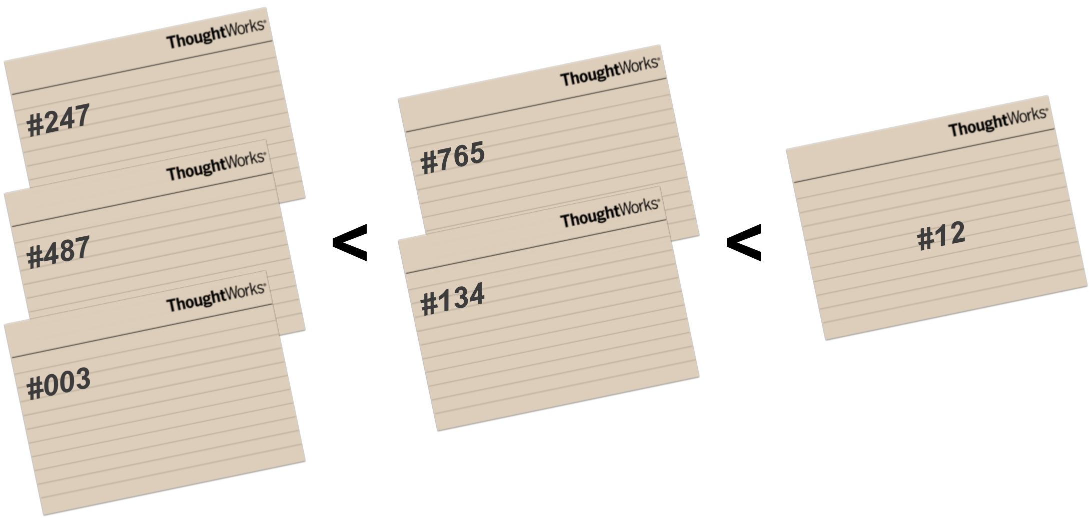
***Distance from the airport can be in another story.***

# INVEST Principle

N Negotiable

# INVEST Principle

## E Estimable



# Getting practical

- **I** Independent
- **N** Negotiable
- **V** Valuable
- **E** Estimable
- **S** Small
- **T** Testable

*Ask the questions in this order*

- **V** Valuable
- **T** Testable
- **S** Small
- **I** Independent
- **N** Negotiable
- **E** Estimable

# What is in a user story

- “As a …  I want … So that …”
- Acceptance Criteria
- Prototype
- User Interface design
- Other text/images/content to provide context

# Acceptance criteria

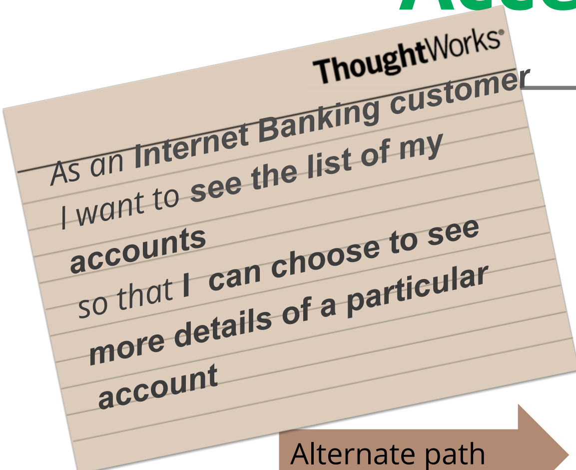
- Tell you the story is finished Acceptance tests tell you the system is working
- A story can have one or multiple AC’s
- It informs the criteria to the tests that will be implemented or executed
- Common format:

  Give **<context>**

  When **<event>**

  Then **<outcome>**

# Acceptance criteria




**Given** the customer has one transaction account and one credit account
**When** they have completed logging in
**Then** the screen should show the names and numbers of the two accounts sorted in account number order

Alternate path

**Given** the customer has just one transaction account
**When** they have completed logging in
**Then** the screen should show the name and number of the account

Alternate path

**Given** the customer has no accounts
**When** they have completed logging in
**Then** the screen should show a message stating that no accounts are available

**Given** the customer has more than 20 accounts
**When** they have completed logging in
**Then** the screen should show the first 20 accounts (in account number order) only

Bad path

**Given** the customer has some accounts
**And** they have completed logging in
**When** the system cannot retrieve the account details
**Then** the screen should show an error message with associated code

# Acceptance criteria

- Tell you the story is finished Acceptance tests tell you the system is working
- A story can have one or multiple AC’s
- It informs the criteria to the tests that will be implemented or executed
- Common format:

  Give **<context>**

  When **<event>**

  Then **<outcome>**

# What is *not* in a user story

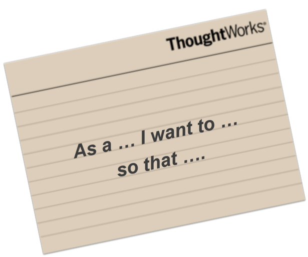
- Unit and Integration tests
- Functional/Acceptance tests
- Documentation – good documents are like vacation pictures
- Specifications

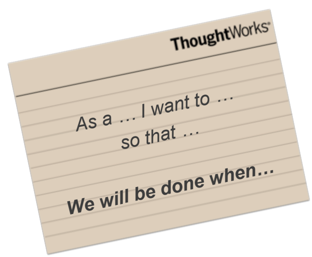
# Story details at the right time

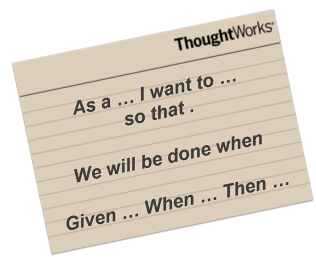
***Vision Workshop***

***Iteration Planning***

***Just-in-time for development***

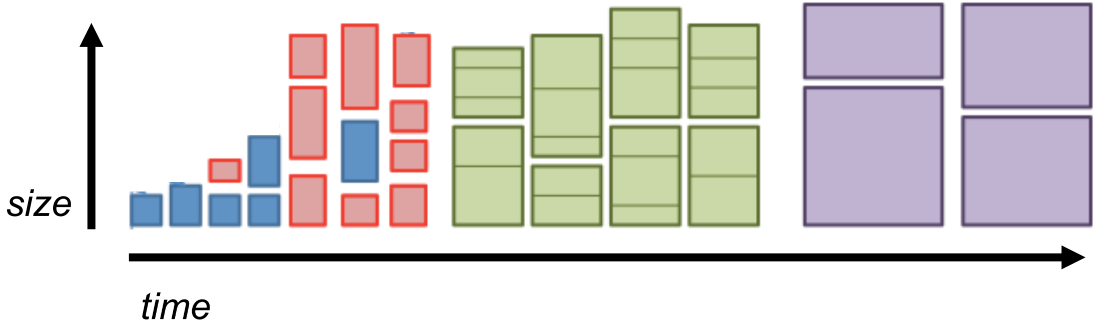


# Why just-in-time?

- Reduces potential wastage
- Provides flexibility to change, prioritize
- Enables learning from delivery
- Tighter feedback loop between business and the delivery team

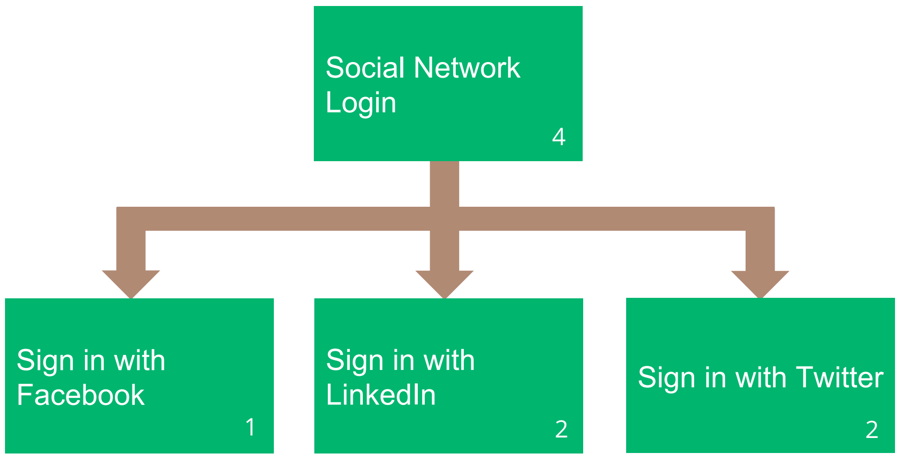


# Story Breakdown

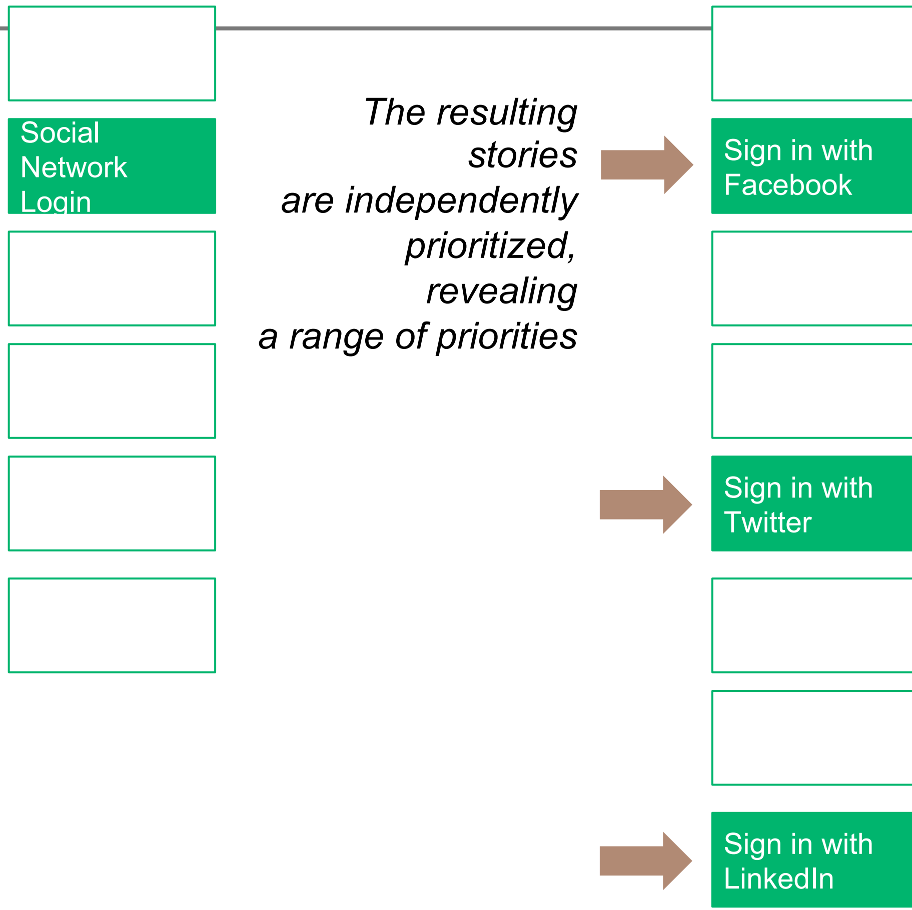
# Story breakdown

- Identify opportunities to split individual stories into one or more smaller independent stories, in order to:
  - Make stories an appropriate size for delivery
    *e.g.. small enough that they are easily understandable, estimable and will give a good indication of progress*
  - Identify differences in priority
    *to ensure only the highest priority work is completed first*

# Splitting by feature



# Splitting by priority



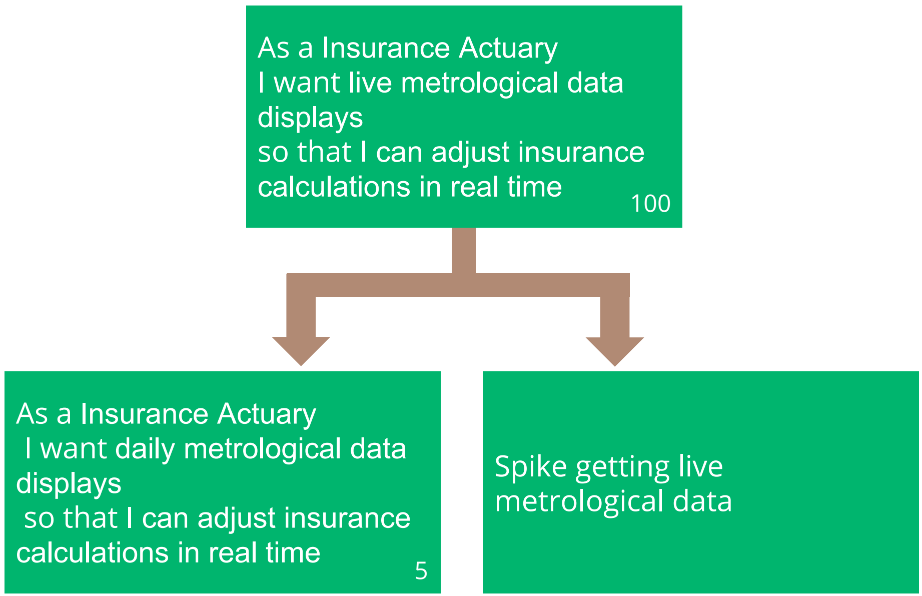
# Splitting by value

As a Call Centre Manager
I want call queue statistics so that I can report on the efficiency of the call centre team
5

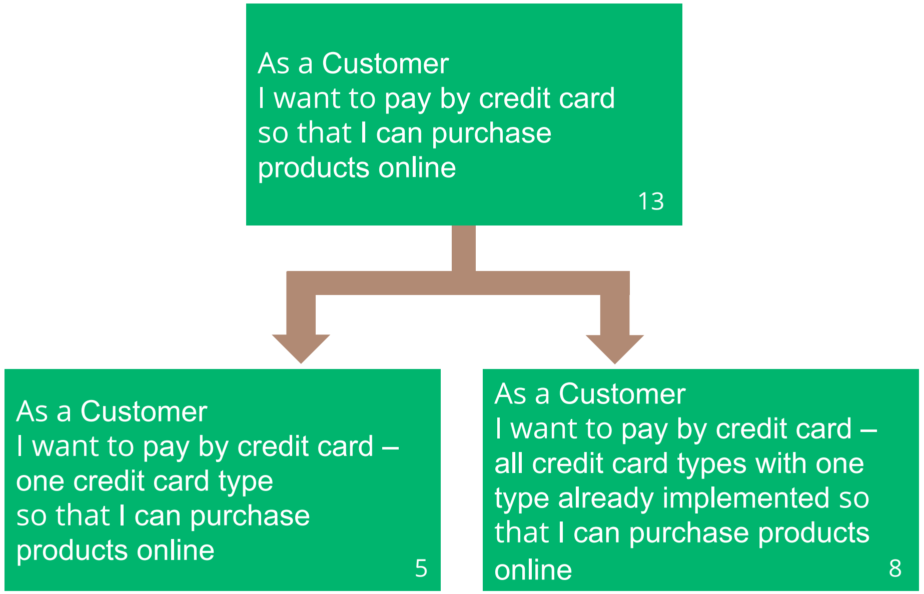
As a Call Centre Manager
I want to see average wait time during peak hours
so that I can staff the call centre adequately
3

As a Call Centre Manager
I want to see average call duration
so that I can report on the efficiency of the call centre team
2

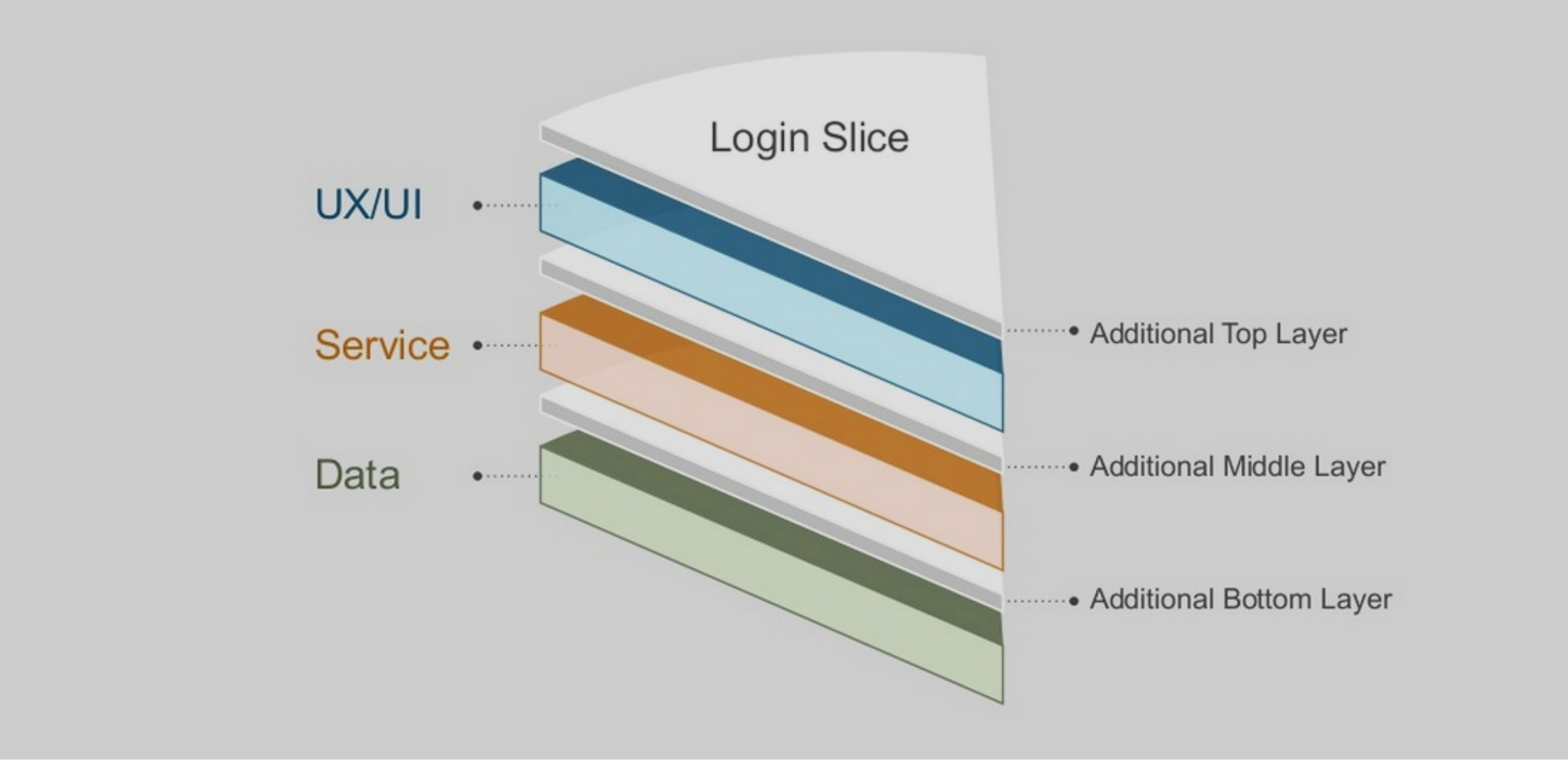
# Splitting by risk



# Splitting dependencies



# Slicing vertically

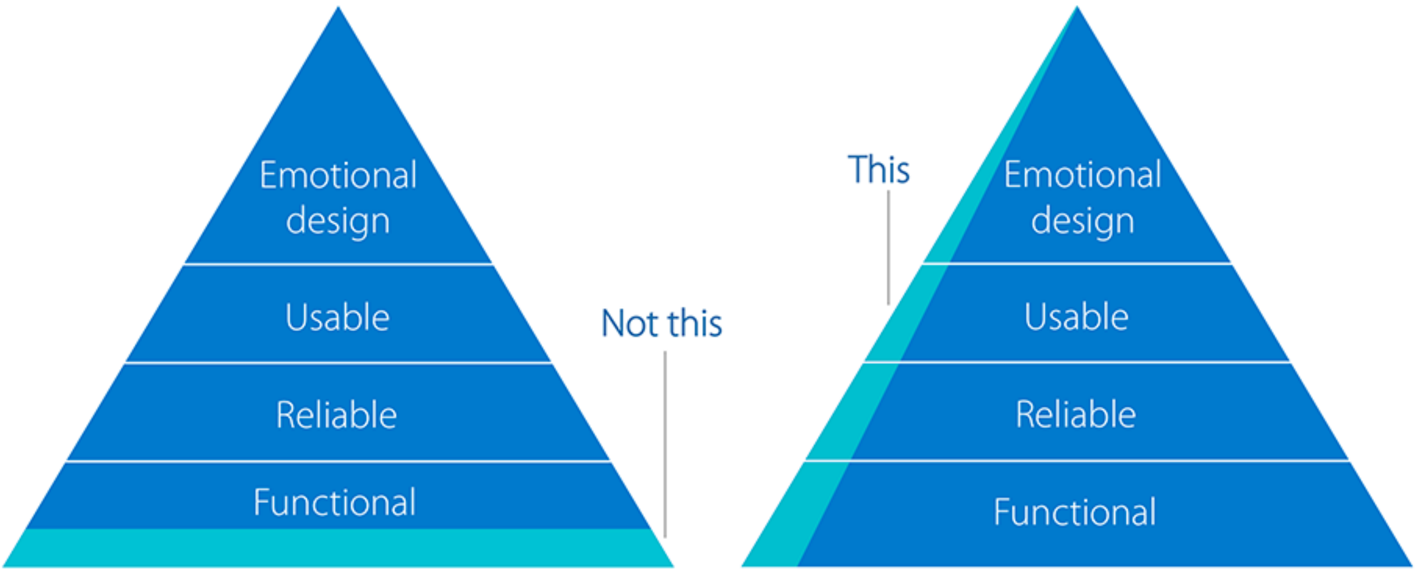



*Made by Ben Clay*

# Slicing vertically


*Made by Ben Clay*

# Minimum Viable Product




@jopas

September 2014 | With compliments to Aarron Walter

# Story Slicing

Slice your stories like you slice your cake...
...Vertical slices across all layers.
This way, when something is done, anyone can see how the slice will fit in the bigger picture

# Things to Remember When Writing User Story

## Six DO's for Writing User Stories

1. Make sure there is a business goal for the story
2. Avoid passive language
3. Narrative should match the Acceptance Criteria
4. Clear Give, When, Then
5. Don't forget the sad path
6. Keep it simple!

## Think about your business user and goal

- User story should be written from the end user perspective
  - helps to keep in mind who the functionality for
  - Focus is on the business goal

# Avoid passive/unclear language

- Use clear direct language when writing the narrative and the acceptance criteria
  - For example:
    - *Should, Could, Usually, Maybe*
    - Unclear wording :
      - There should be an error response
      - Should return the details from before...

# Narrative = AC

- Any details of what should happen as part of a user story should also be reflected in the Acceptance Criteria
- This is important to make sure that all details and functionality that are called out are addressed, implemented and testable

# Clear ACs

- Clear concise statements
- Avoid multiple <WHEN>s in one AC
- Avoid multiple <THEN>s in one AC

# Don’t forget Sad Path

- A lot of times when writing a story you get focused on how the story should work = Happy Path
- A story is not ‘complete’ until you address all the possible scenarios
  - What happens if there is an error in the input?
  - What happens if the service is unavailable?
    - Asking questions like this can also open up discussion and/or underlying NFR needs

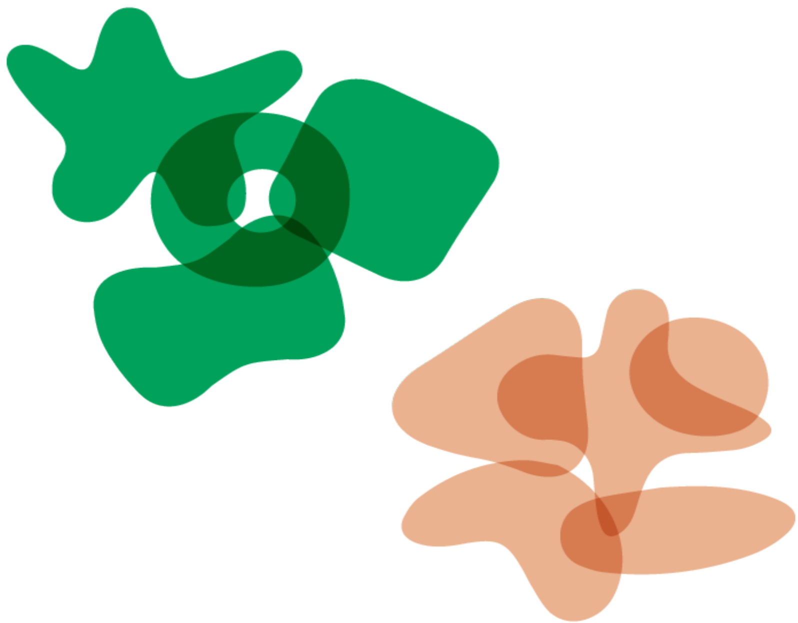
# Keep it Simple

- A lot of times when writing a story you get focused on how the story should work = Happy Path
- A story is not ‘complete’ until you address all the possible scenarios
    - What happens if there is an error in the input?
    - What happens if the service is unavailable?
        - Asking questions like this can also open up discussion and/or underlying NFR needs

# This is not a static template

- Keep that as the project and team changes so will the user stories
- Information that was needed at one point may not be necessary any longer
- What information is most important to see in a user story?
  - Where do you focus? The whole card? Only the AC? Is the Background important to up?
- Goal is to have anyone (in or outside the team) be able to read a story card and understand what it is about

# You've Got Your Stories, How Do They Fit Together?



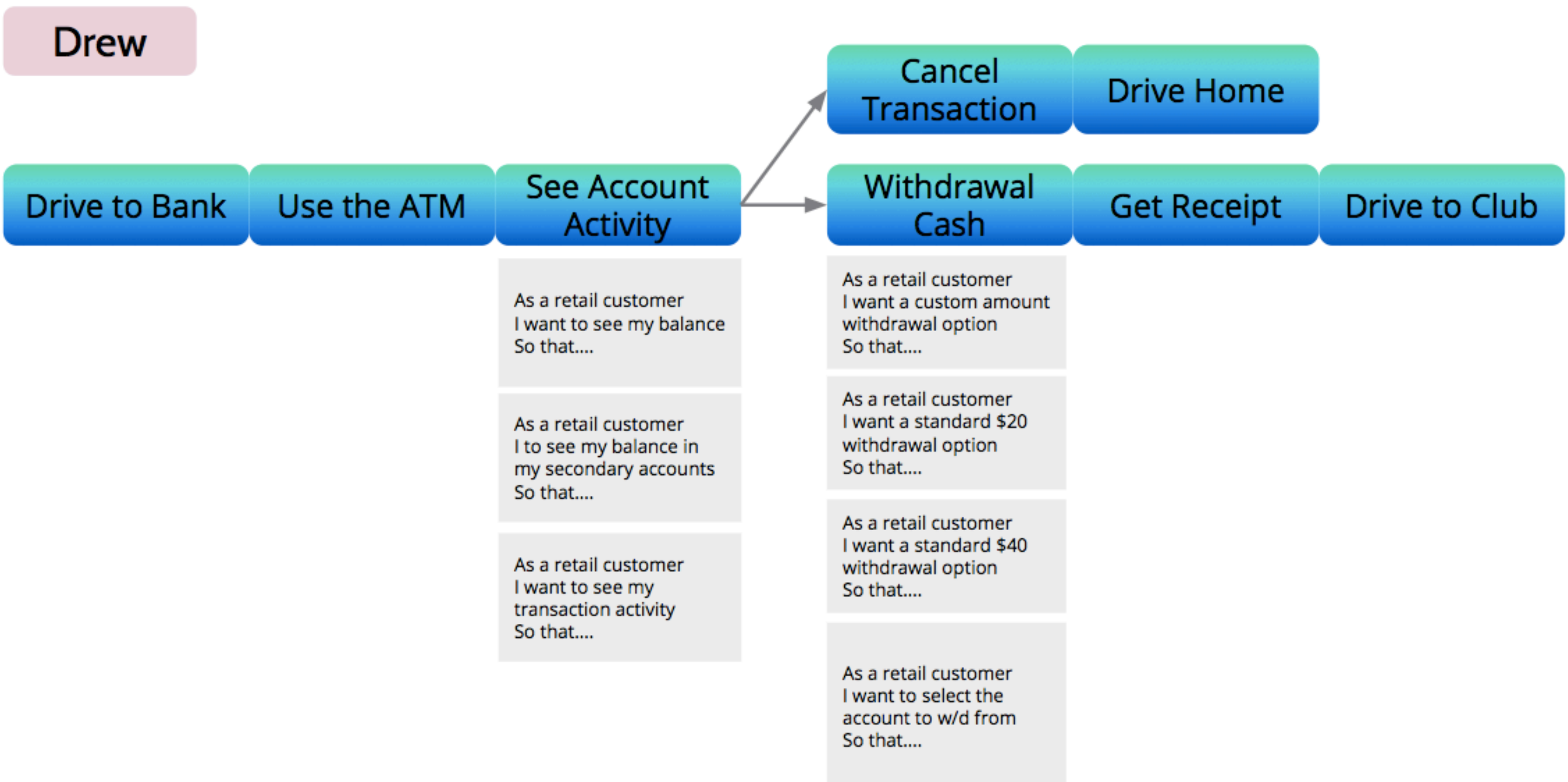
# What is Story Mapping?

A user story map arranges user stories into a useful model to help you:

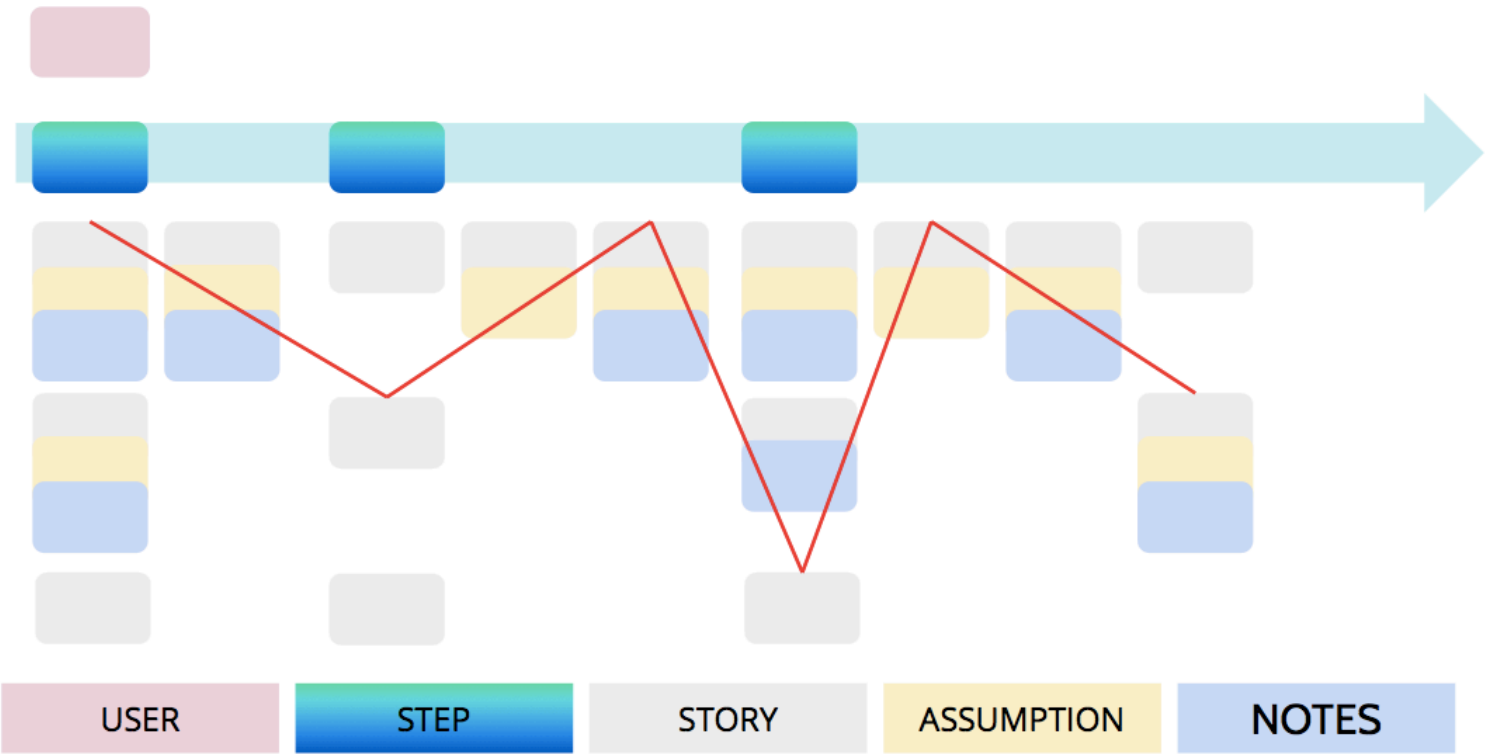
- ...understand the functionality of the system.
- ...identify holes and omissions in your backlog.
- ...effectively prioritize and groom your backlog so that

value is delivered every iteration.

# Map it Out


Drew
Cancel Transaction
Drive Home
Drive to Bank
Use the ATM
See Account Activity
Withdrawal Cash
Get Receipt
Drive to Club
As a retail customer
I want to see my balance
So that....
As a retail customer
I to see my balance in
my secondary accounts
So that....
As a retail customer
I want to see my
transaction activity
So that....
As a retail customer
I want a custom amount
withdrawal option
So that....
As a retail customer
I want a standard $20
withdrawal option
So that....
As a retail customer
I want a standard $40
withdrawal option
So that....
As a retail customer
I want to select the
account to w/d from
So that....

# Prioritize Based On Necessity



# ORGANIZE BY PRIORITY

Drew

PRIORITY

## See Account Activity

As a retail customer
I want to see my balance
So that....

As a retail customer
I want to see my transaction activity
So that....

As a retail customer
I to see my balance in my secondary accounts
So that....

## Withdrawal Cash

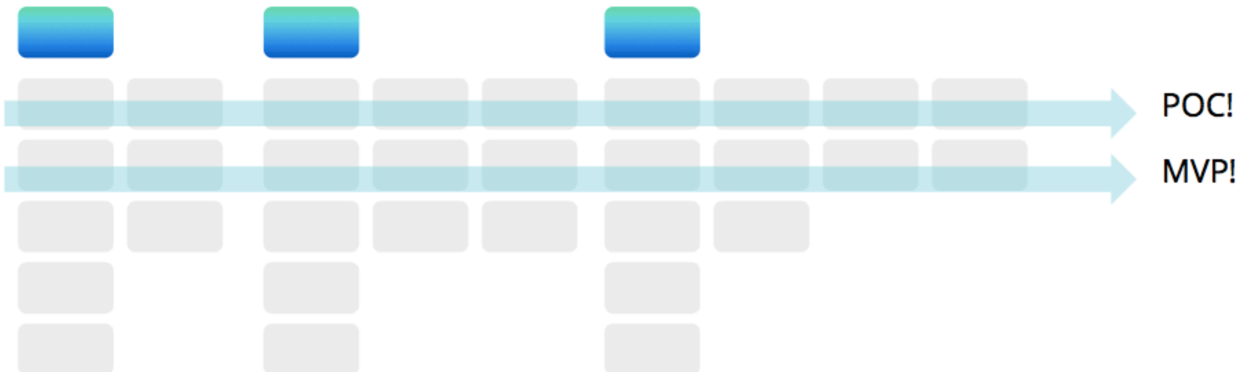
As a retail customer
I want a custom amount withdrawal option
So that....

As a retail customer
I want to select the account to w/d from
So that....

As a retail customer
I want a standard $20 withdrawal option
So that....

As a retail customer
I want a standard $40 withdrawal option
So that....

# WALK THE MAP



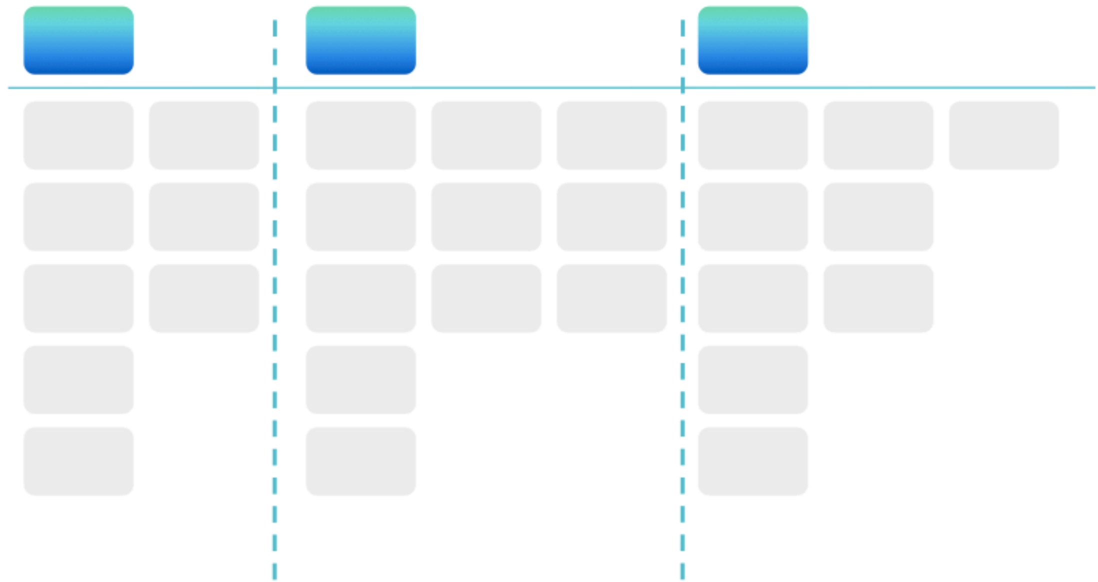

- Walking the map helps to determine your product backlog.
- Do highest value, highest priority items first.

**STORY MAPS ARE DYNAMIC.**

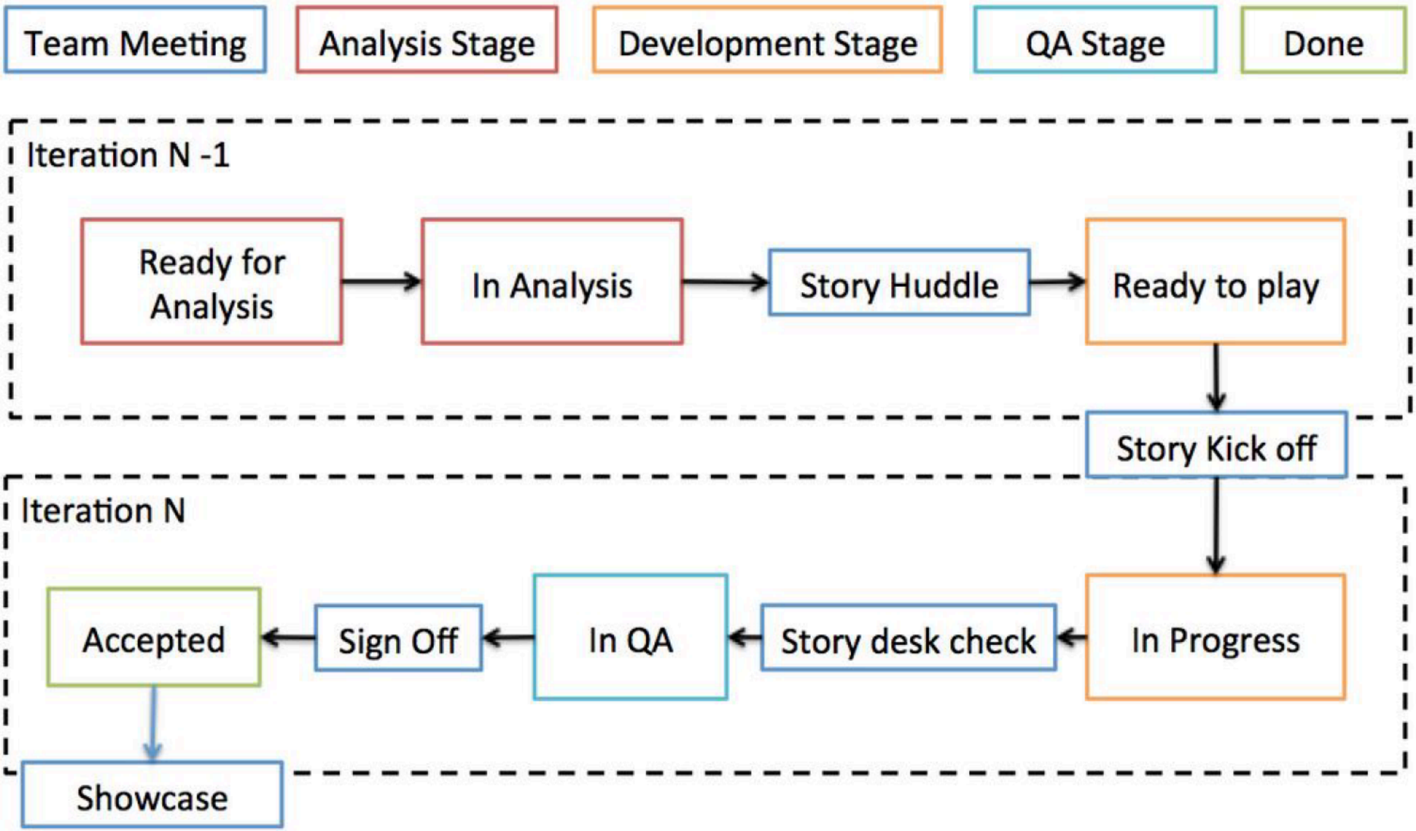
# HEY! A BACKLOG!

FLAT BACKLOG

VS.



# USER STORY LIFECYCLE



# Questions?

ThoughtWorks®

# THANK YOU